

Housing Price Prediction – Project Summary

This project focused on predicting house prices using machine learning techniques on a housing dataset containing 545 records and 13 features. Data exploration showed that the dataset had no missing values and required only categorical feature encoding before model training. Important variables included area, bedrooms, bathrooms, stories, parking availability, air conditioning, preferred area, and furnishing status.

After preprocessing, the data was split into training (80%) and testing (20%) sets. Two regression models were developed and compared: Linear Regression and Random Forest Regressor. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Linear Regression achieved an R^2 score of approximately 0.65, indicating that it explained around 65% of the variation in house prices. Surprisingly, it performed slightly better than the Random Forest model, suggesting that the relationship between the features and house price is largely linear.

Visual analysis through histograms, correlation heatmaps, scatter plots, and feature importance charts revealed that house area is the strongest predictor of price, followed by bathrooms, parking spaces, air conditioning, and number of stories. Larger and better-equipped houses consistently commanded higher prices.

Based on these findings, a real estate business should prioritize marketing properties with larger areas, modern amenities, and adequate parking facilities, as these features contribute significantly to higher property values and buyer demand.